

Decision-Rights Allocation: A Governance-Centric Perspective on LLM-Enabled Agentic Systems

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Abstract

LLM-enabled agentic systems are increasingly deployed inside task-and-decision processes that retrieve information, reason over context, call tools, coordinate with other agents, and execute actions on behalf of humans. Existing conceptual work has usefully classified these systems by external behavior, architectural components, cognitive design patterns, autonomy levels, and multi-agent governance forms. Those frameworks clarify what agentic systems contain and how they behave, but they often leave implicit a design choice that becomes central once stochastic language models are connected to organizational processes: which party has the effective right to make each consequential decision? This paper proposes decision-rights allocation as a compact internal-mechanism perspective for understanding and designing LLM-enabled agentic systems. We define an agentic system as an allocation of decision rights among three parties: human actors, deterministic components, and stochastic LLM components. The distinction matters because LLMs do not merely add another software module; they introduce open-ended semantic judgment into information selection, reasoning, action selection, validation, and memory update. The framework interprets common design patterns—including retrieval-augmented generation, episodic memory, human-in-the-loop validation, domain-expert knowledge codification, provenance tracing, reinforcement learning, routing, schema validation, and ontology-derived semantic context—as mechanisms that shift decision rights across these three parties. We further propose a simple optimization model in which the desirable allocation depends on task risk, adaptability needs, scalability pressure, auditability requirements, deterministic codifiability, human capacity, model reliability, verification availability, and the cost of errors at scale. The contribution is a unifying vocabulary for comparing agentic-system designs and a first-step analytical model for deciding when LLMs should decide, when deterministic infrastructure should constrain them, and when humans should retain or regain control.

Introduction

Artificial Intelligence (AI) agent is not a new concept [1], [2], [3], [4], [5]. What distinguishes an LLM-based AI agent (LLM agent hereafter) from its predecessors is that its pre-trained model can semantically understand unstructured data, particularly in the form of human natural language [6], [7]. LLM agents thus show much higher adaptability and scalability compared to rule-based symbolic AI agents, as LLMs can evolve through post-training or in-context learning driven by natural language data [8], [9], [10], [11]. This strength has contributed to the proliferation of LLM agent research and applications in recent years.¹

LLM agents, however, also have their adversaries. Greater monetary costs and longer runtimes are two challenges described by LLM agent surveys [11], [12]. In addition, users’ trust in LLM agents could be eroded by a lack of transparency [13], stochastic features [14], and unawareness of uncertainty [15]. When

¹According to statistics from “<https://cognaptus.com/datahub/ai-research-trends/>”, before 2024, the share of agentic AI-related papers accounted for less than 3% of cs.AI papers on arXiv; in 2025, this share increased to 9.2%, and in the first half of 2026, it further increased to 11.2%.

we introduce modules such as perception, memory, reflection, and communication with other LLM agents in practical LLM-enabled use cases, an LLM agent becomes an LLM-enabled agentic AI system (LLM agentic system hereafter), whose complexity further worsens the trust problem [16], [17], [18]. As a result, doubts about LLM agents persist [19], [20], [21], [22].

This paper proposes that a good design for an LLM agentic system should be based on the decision-node level. Particularly, this paper proposes that the fundamental design issue at the decision-node level is how to allocate decision rights among three types of actors: humans, deterministic rule-based mechanisms, and LLMs. We classify these three types of actors not only to simplify our model, but also because they are quite different in those features that represent the strengths and weaknesses of LLMs. <Here I am still thinking about their differences; human actors may be more expensive and harder to scale, but very adaptable and more reliable in certain aspects; deterministic components are cheap and also scalable, but not adaptable.>

This paper not only raises this concept, but also proposes a paradigm to optimize the node-level decision-right allocation problem. We first suggest a macro-level architectural linear classifier model to estimate the initial decision-right allocation status. It is an architectural model because we first use node-level metrics (the sub-task represented by the node) to estimate latent variables such as demands for transparency, demands for fast runtime, demands for non-stochastic results, and demands for uncertainty awareness. Based on those latent variables, we further determine the decision-right allocation vector for that node.

Then we let the LLM agentic system evolve iteratively with real data. The evolution should take into account financial costs, customer feedback, extremely bad outcomes, etc. The evolution adjusts all node-level decision-right allocation vectors jointly because they can propagate through or relate to each other.

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